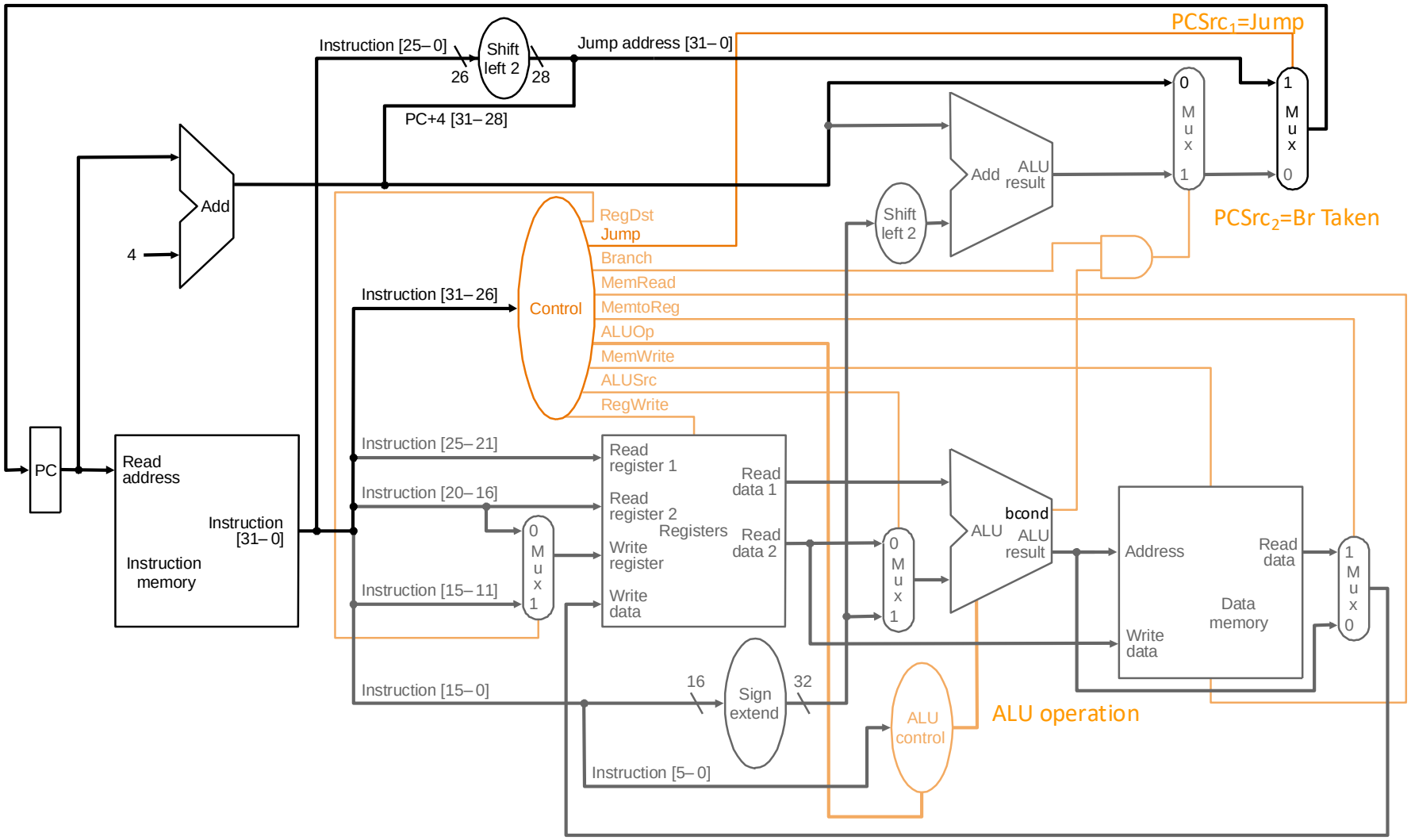
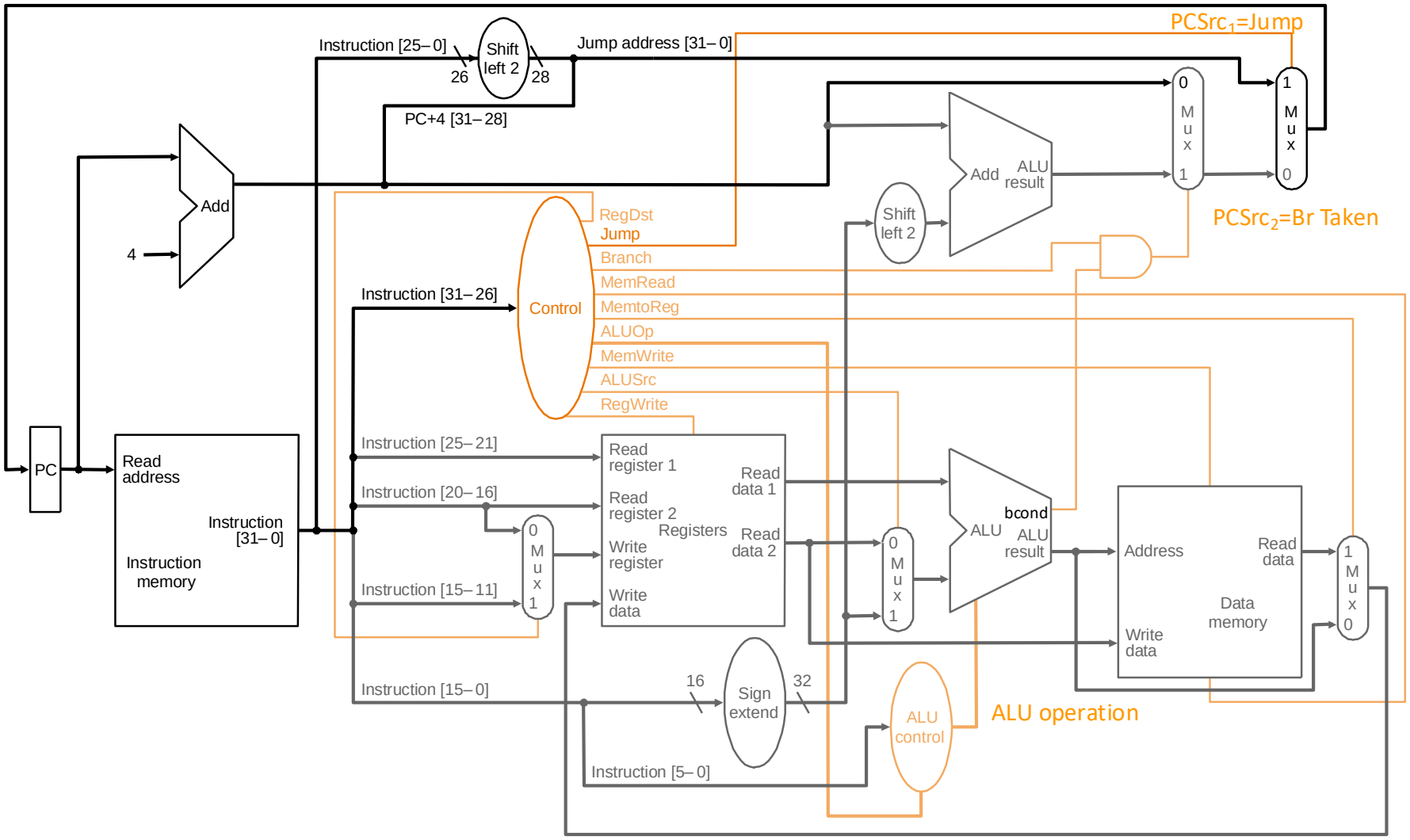
Putting It All Together



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JAL, JR, JALR omitted

Generate Control Signals (in Orange Color)



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JAL, JR, JALR omitted

Single-Bit Control Signals (I)

	When De-asserted	When asserted	Equation
RegDst	GPR write select according to **rt**, i.e., inst[20:16]	GPR write select according to **rd**, i.e., inst[15:11]	opcode==0
ALUSrc	2nd ALU input from 2nd **GPR** read port	2nd ALU input from sign-extended 16-bit **immediate**	(opcode!=0) && (opcode!=BEQ) **&&** (opcode!=BNE)
MemtoReg	Steer **ALU result** to GPR write port	Steer **memory output** to GPR write port	opcode==LW
RegWrite	GPR **write disabled**	GPR **write enabled**	(opcode!=SW) && (opcode!=Bxx) && (opcode!=J) && (opcode!=JR))

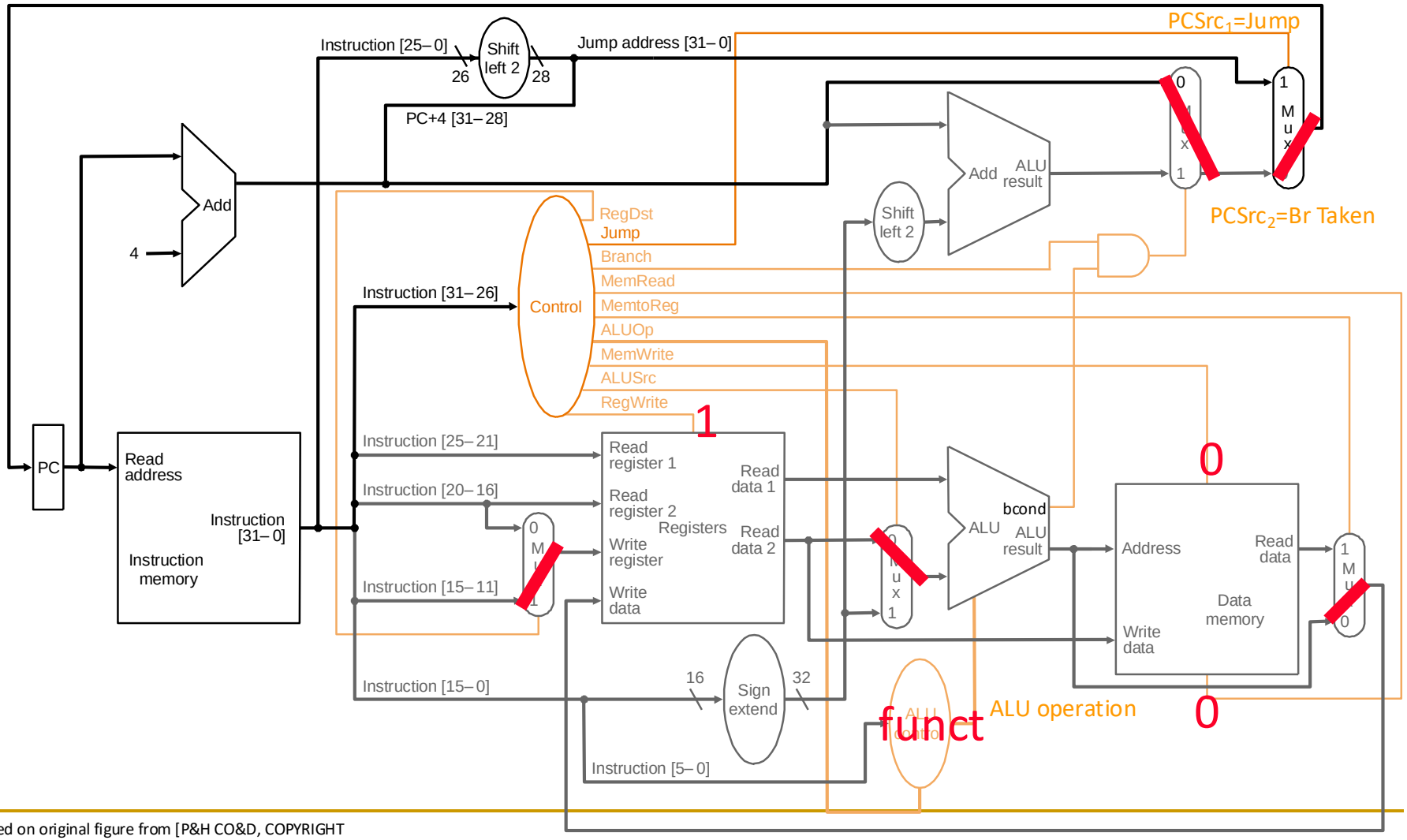
JAL and JALR require additional RegDst and MemtoReg options

Single-Bit Control Signals (II)

	When De-asserted	When asserted	Equation
MemRead	Memory **read disabled**	Memory **read port returns load value**	opcode==LW
MemWrite	Memory **write disabled**	Memory **write enabled**	opcode==SW
$PCSrc_1$	According to **$PCSrc_2$**	next PC is based on **26-bit immediate jump target**	(opcode==J) || (opcode==JAL)
$PCSrc_2$	next PC = **PC + 4**	next PC is based on **16-bit immediate branch target**	(opcode==Bxx) && "bcond is satisfied"

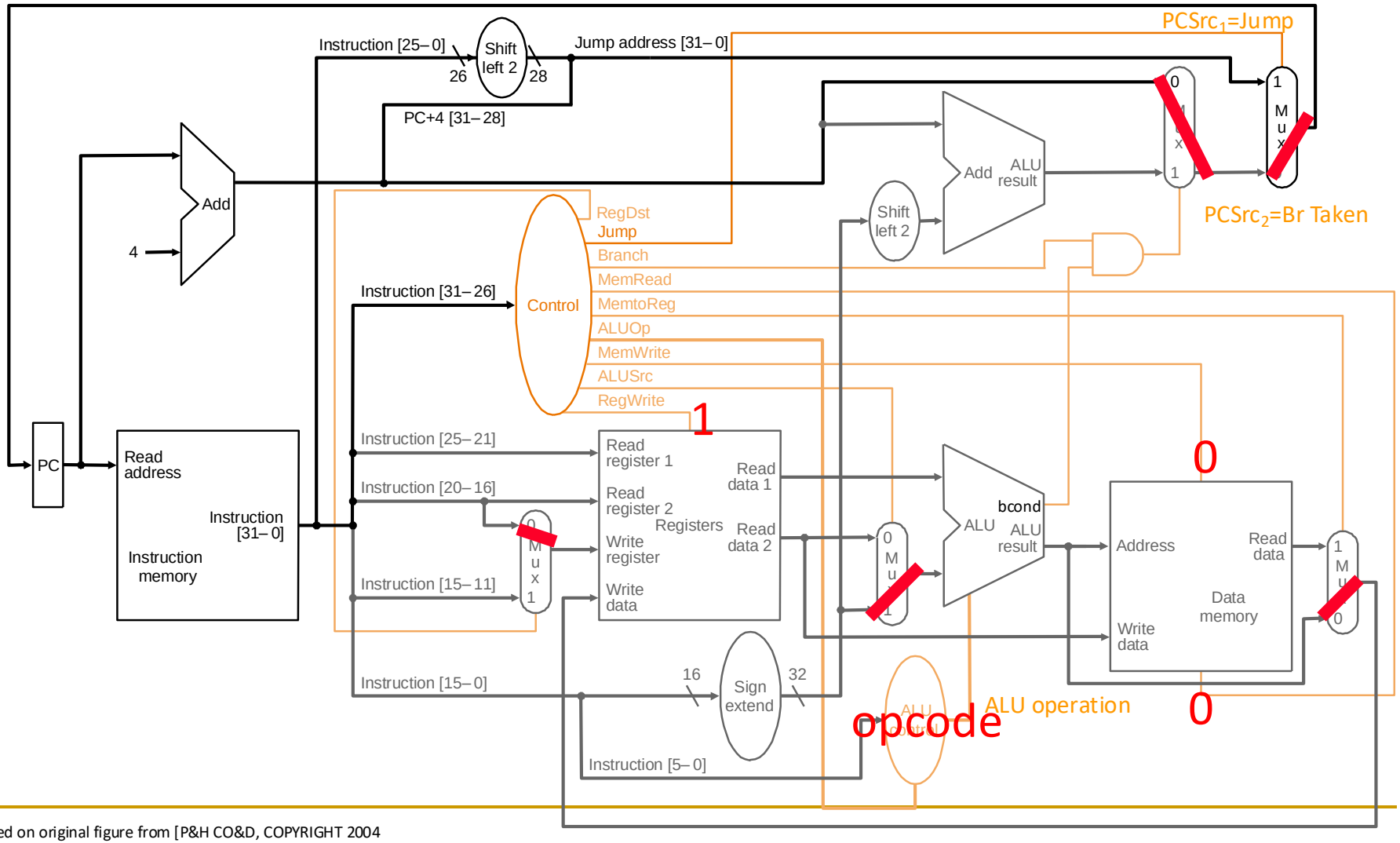
JR and JALR require additional PCSrc options

R-Type ALU



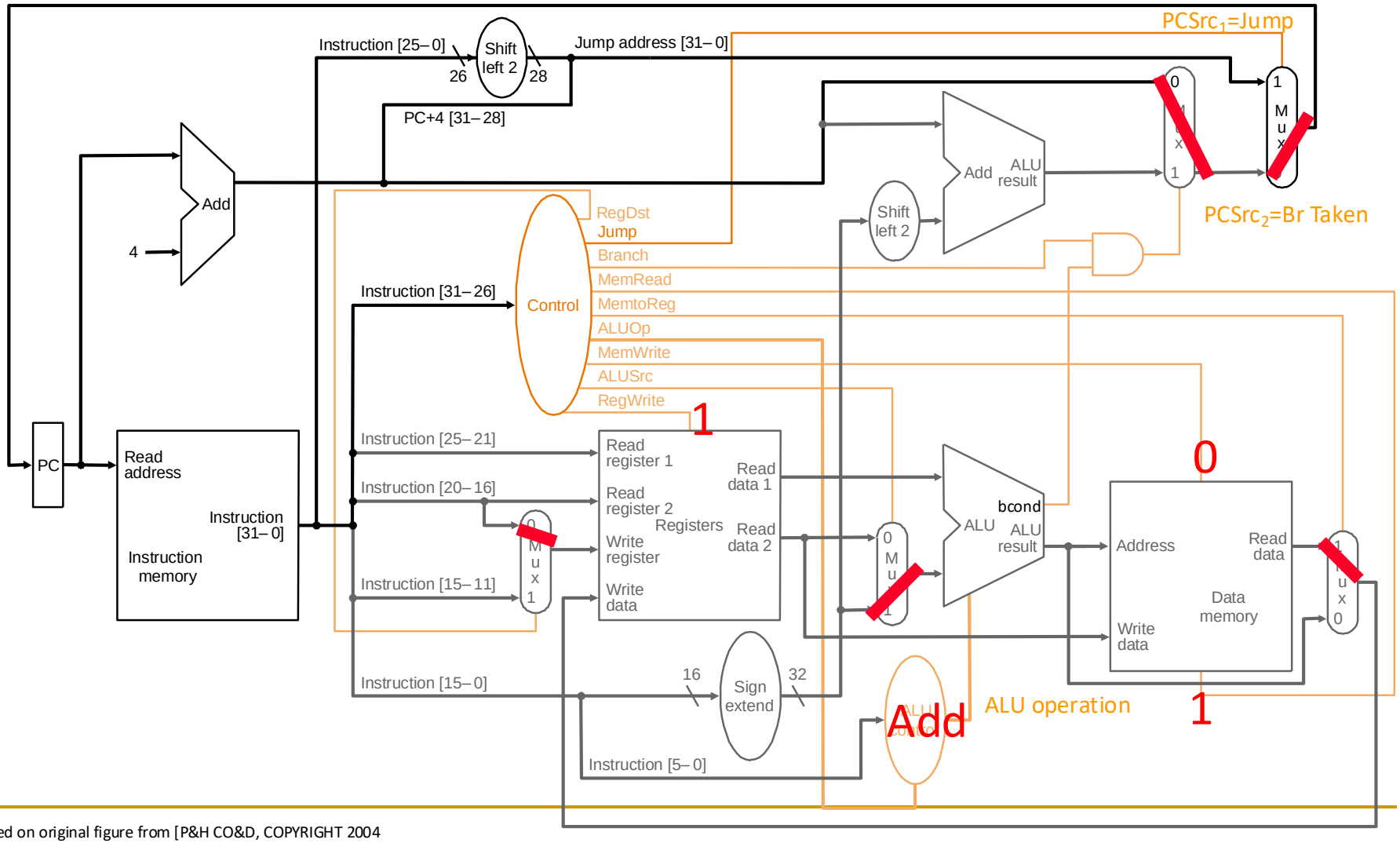
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I-Type ALU



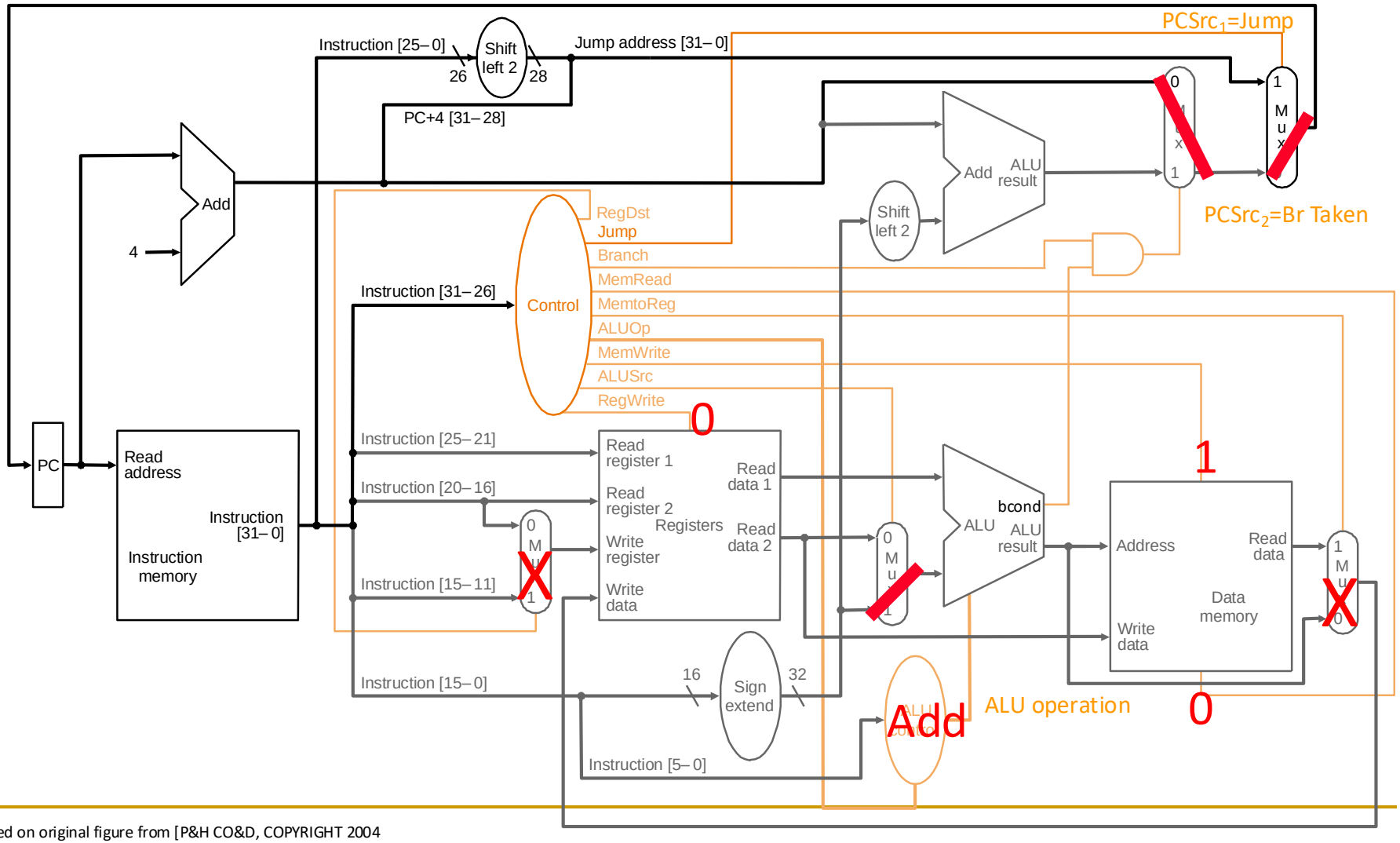
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LW



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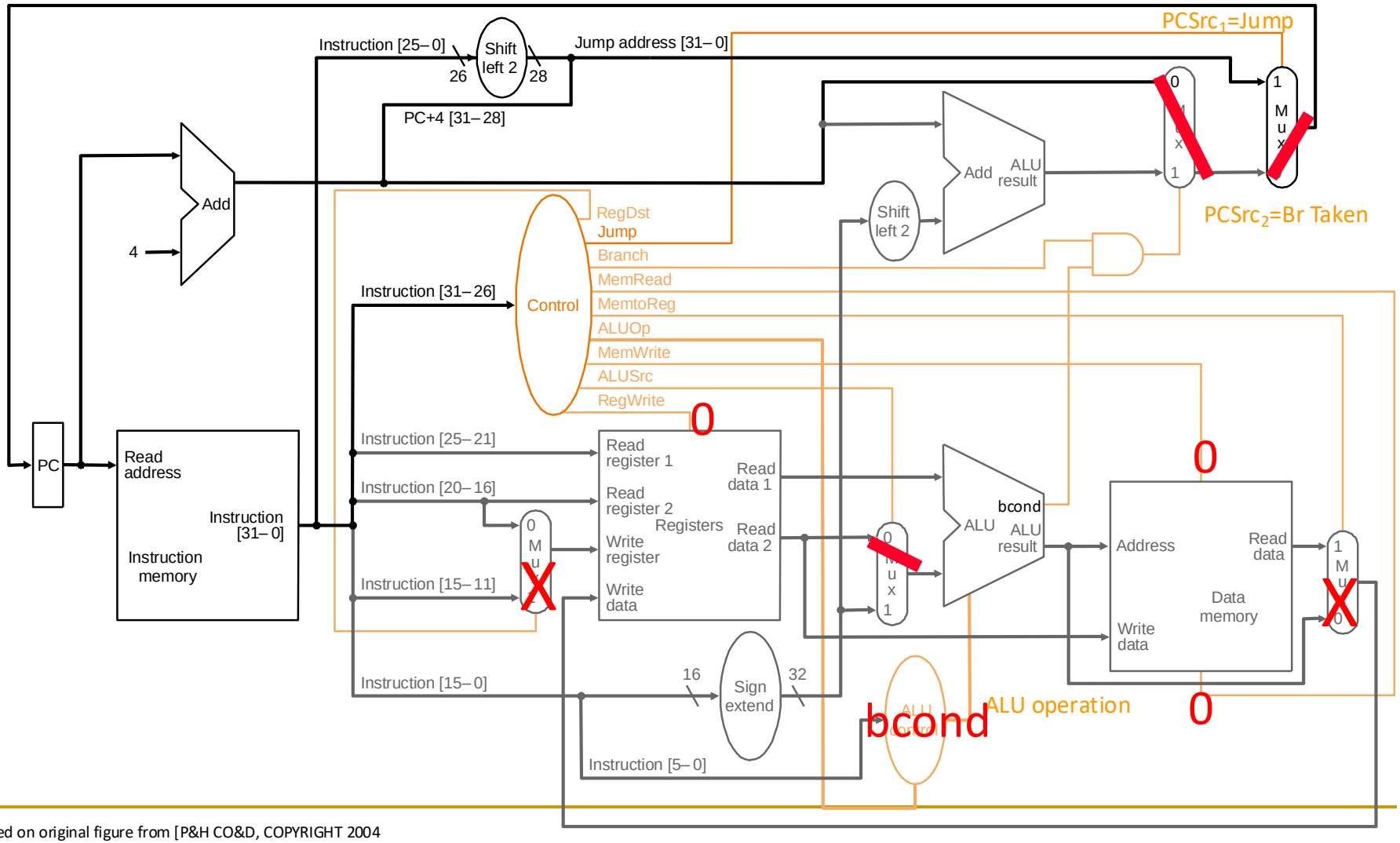
SW



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Branch (Not Taken)

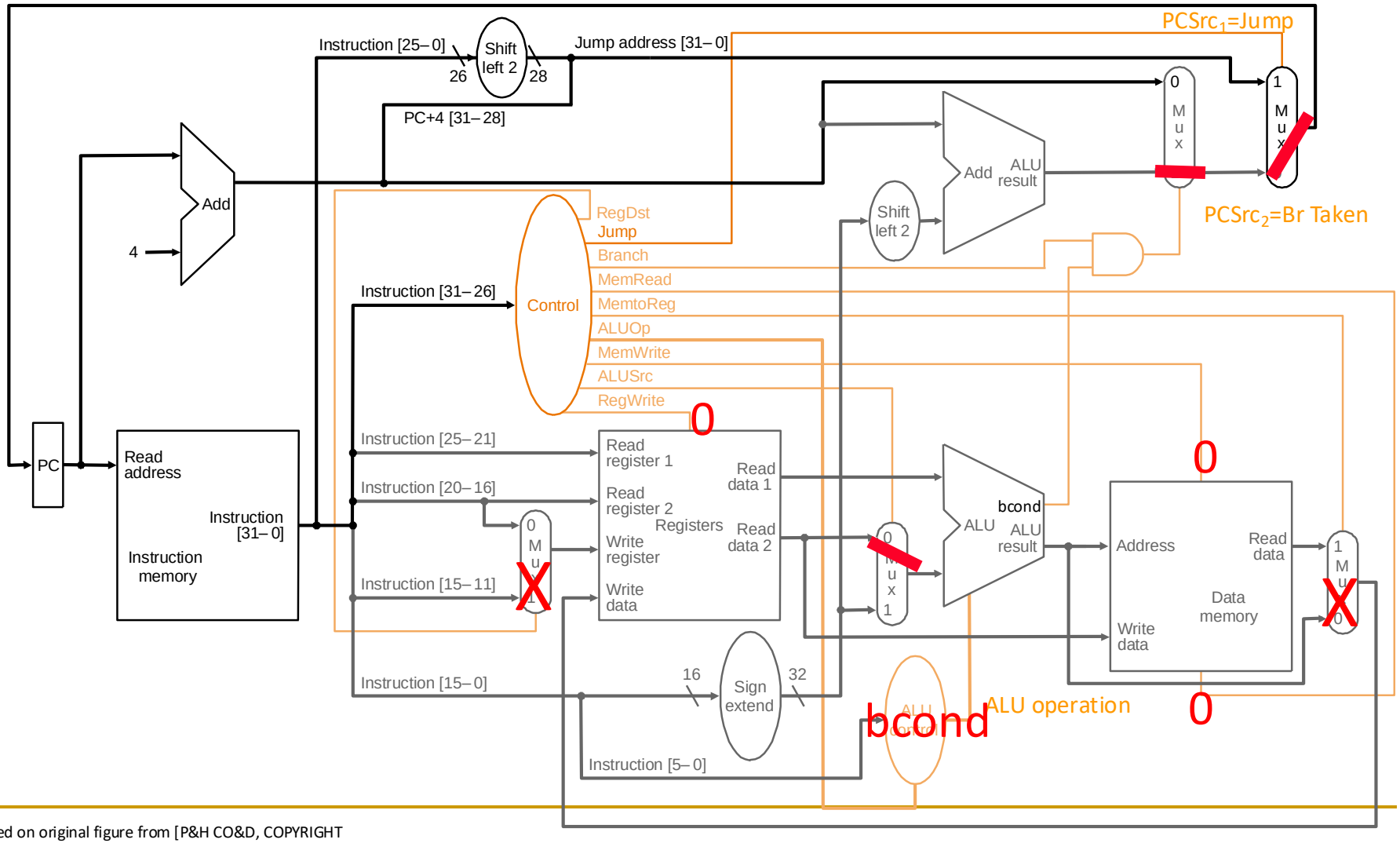
Some control signals are dependent on the processing of data



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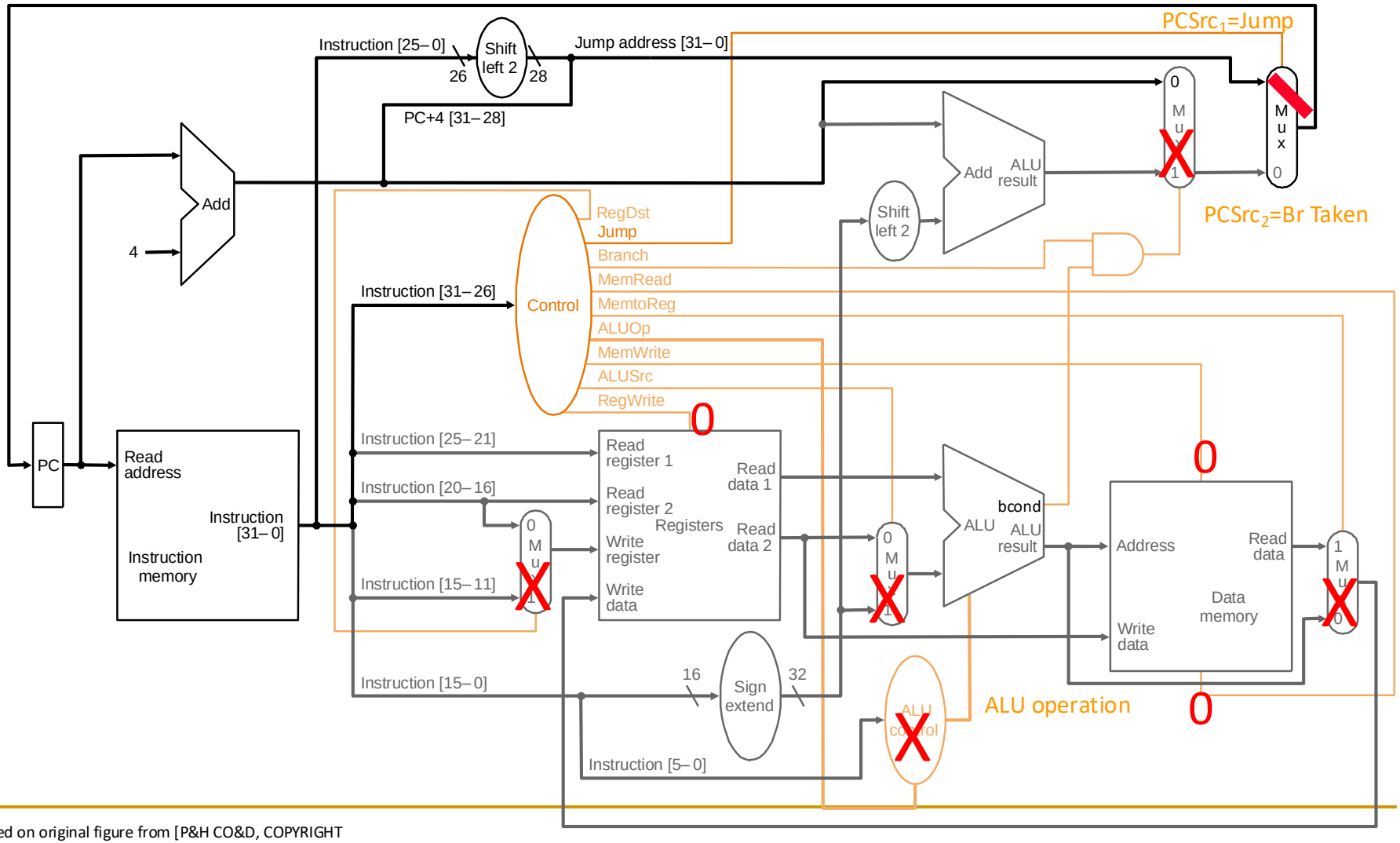
Branch (Taken)

Some control signals are dependent on the processing of data



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Jump



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A Single-Cycle Microarchitecture: Analysis

- Every instruction takes 1 cycle to execute
  - CPI (Cycles per instruction) is strictly 1

- How long each instruction takes is determined by how long the slowest instruction takes to execute
  - Even though many instructions do not need that long to execute

- Clock cycle time of the microarchitecture is determined by how long it takes to complete the slowest instruction
  - Critical path of the design is determined by the processing time of the slowest instruction

What is the Slowest Instruction to Process?

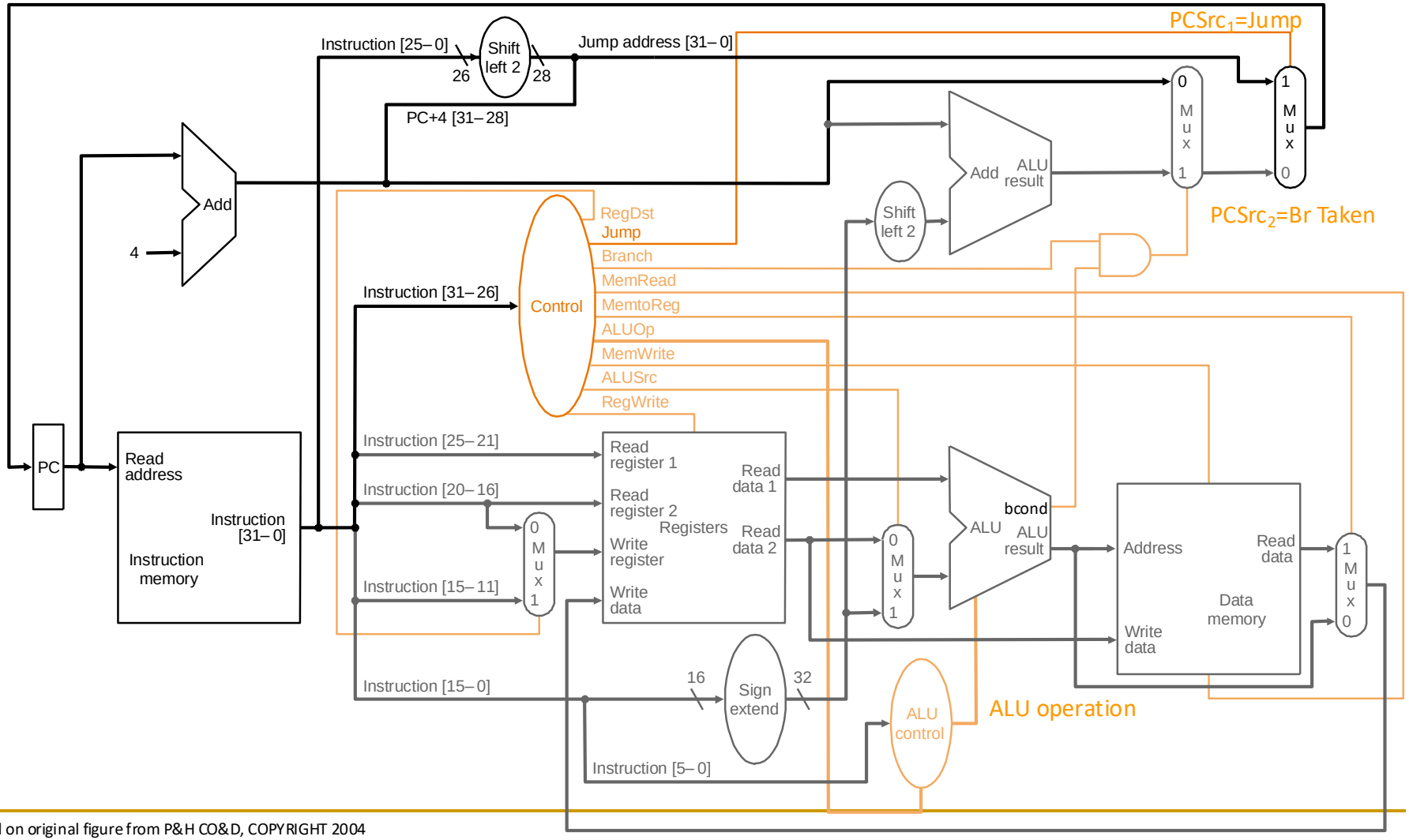
- Let's go back to the basics

- All six phases of the instruction processing cycle take a *single machine clock cycle* to complete
  - Fetch
  - Decode
  - Evaluate Address
  - Fetch Operands
  - Execute
  - Store Result

1. Instruction fetch (IF)
2. Instruction decode and register operand fetch (ID/RF)
3. Execute/Evaluate memory address (EX/AG)
4. Memory operand fetch (MEM)
5. Store/writeback result (WB)

- Does every instruction take the same time (latency) to complete?

Let's Find the Critical Path



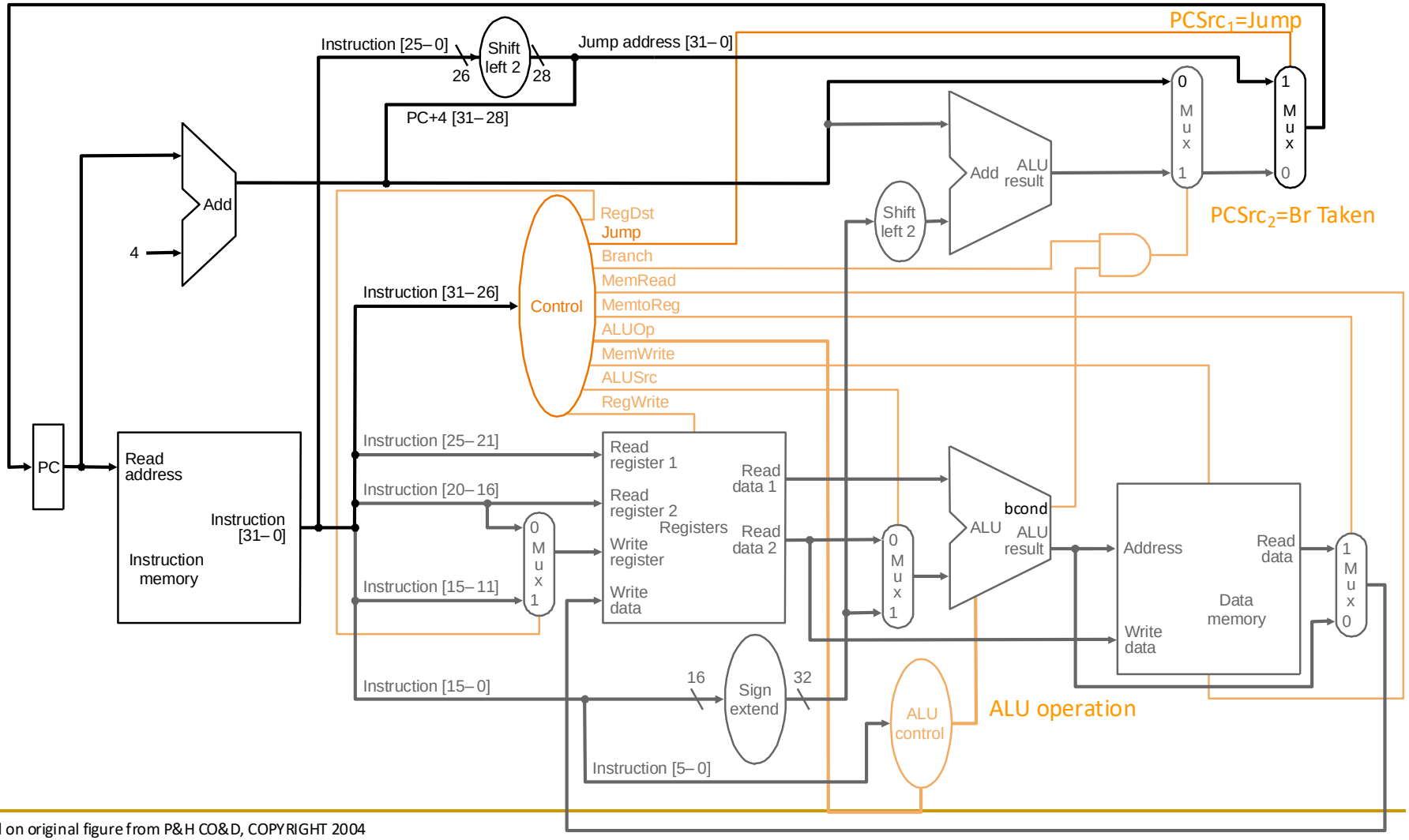
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Example Single-Cycle Datapath Analysis

- Assume (for the design in the previous slide)
  - memory units (read or write): 200 ps
  - ALU and adders: 100 ps
  - register file (read or write): 50 ps
  - other logic or wire delay: 0 ps

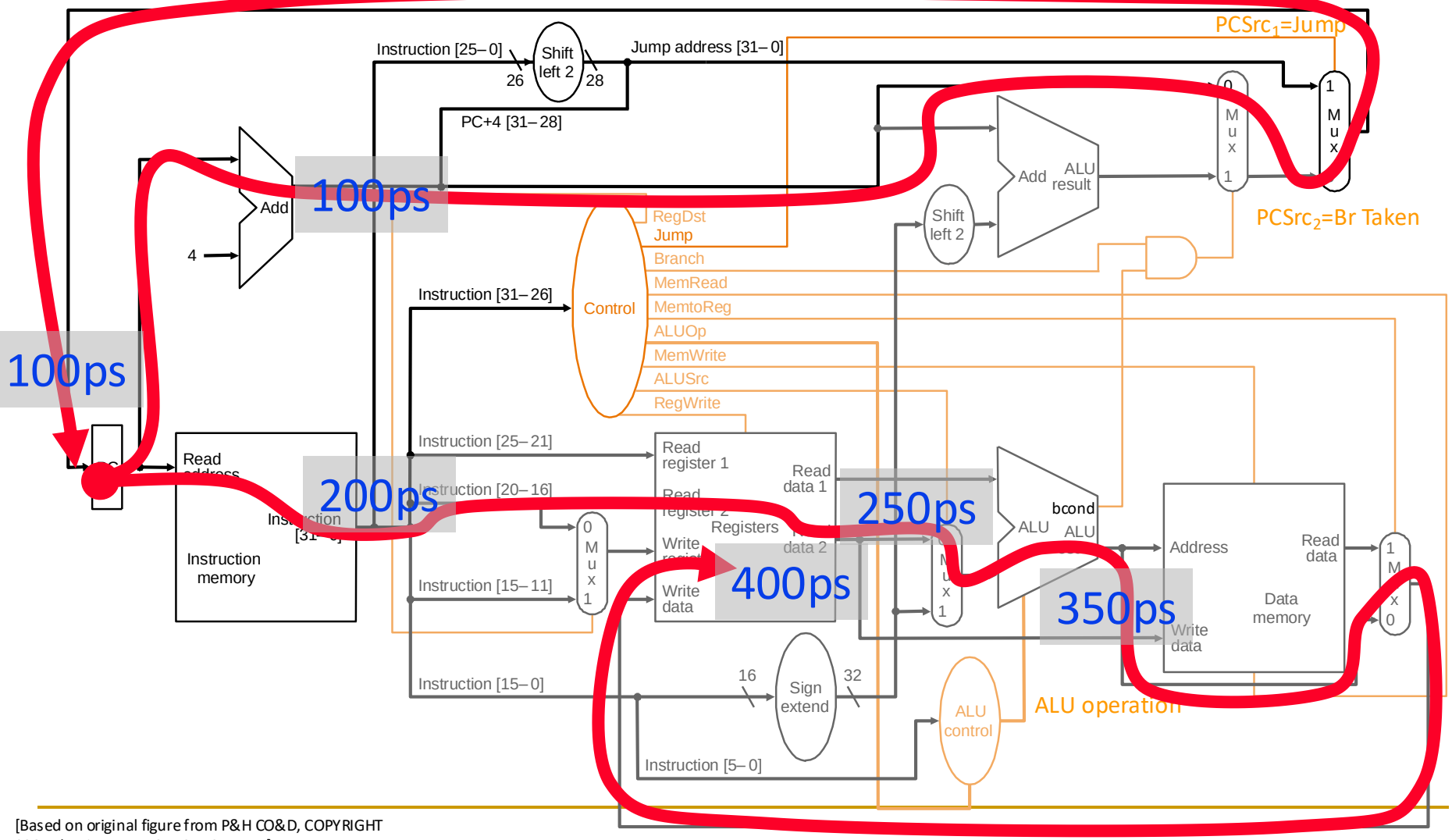
steps	IF	ID	EX	MEM	WB	Delay
resources	mem	RF	ALU	mem	RF	
R-type	200	50	100		50	400
I-type	200	50	100		50	400
LW	200	50	100	200	50	600
SW	200	50	100	200		550
Branch	200	50	100			350
Jump	200					200

Let's Find the Critical Path



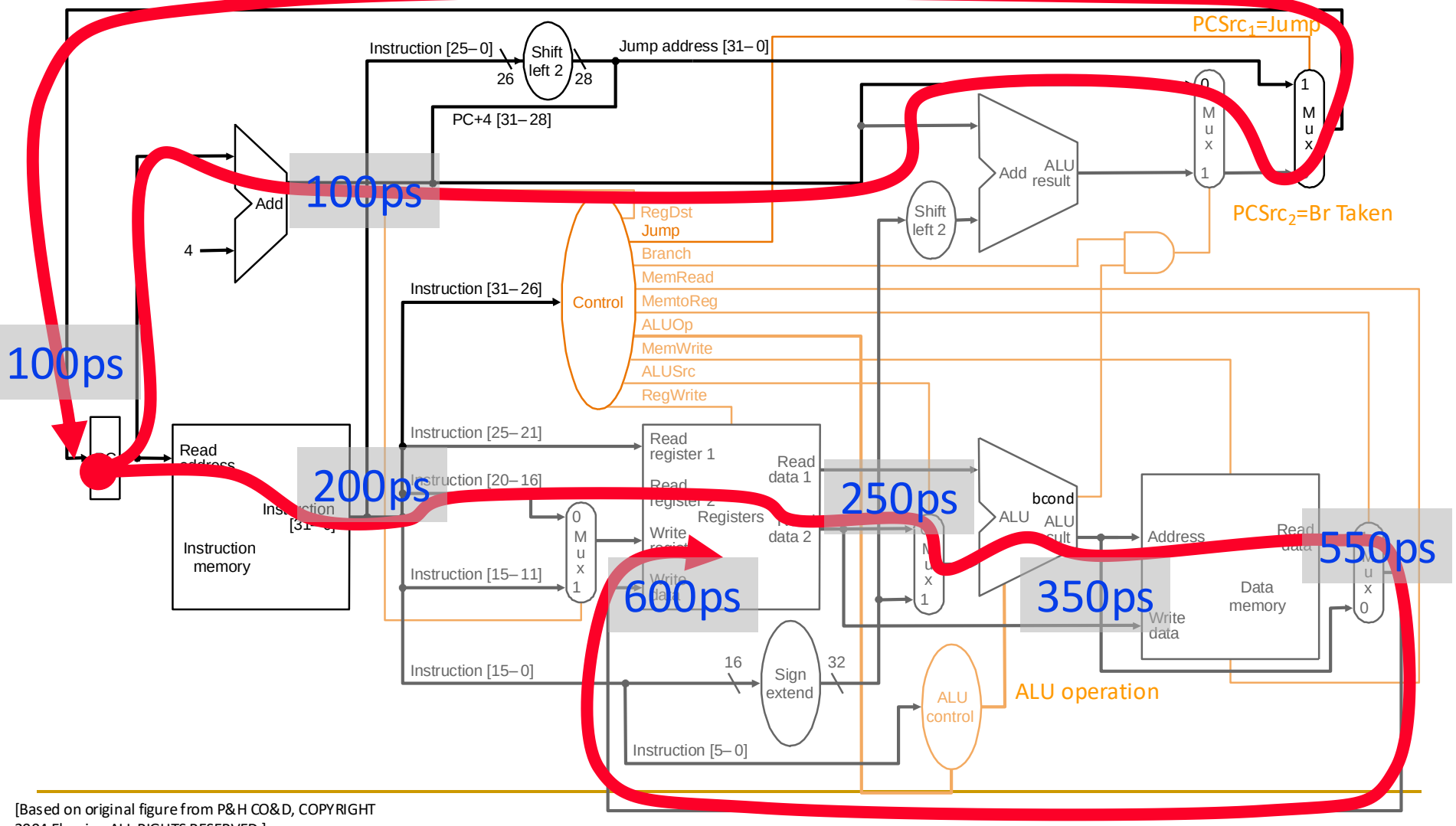
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R-Type and I-Type ALU



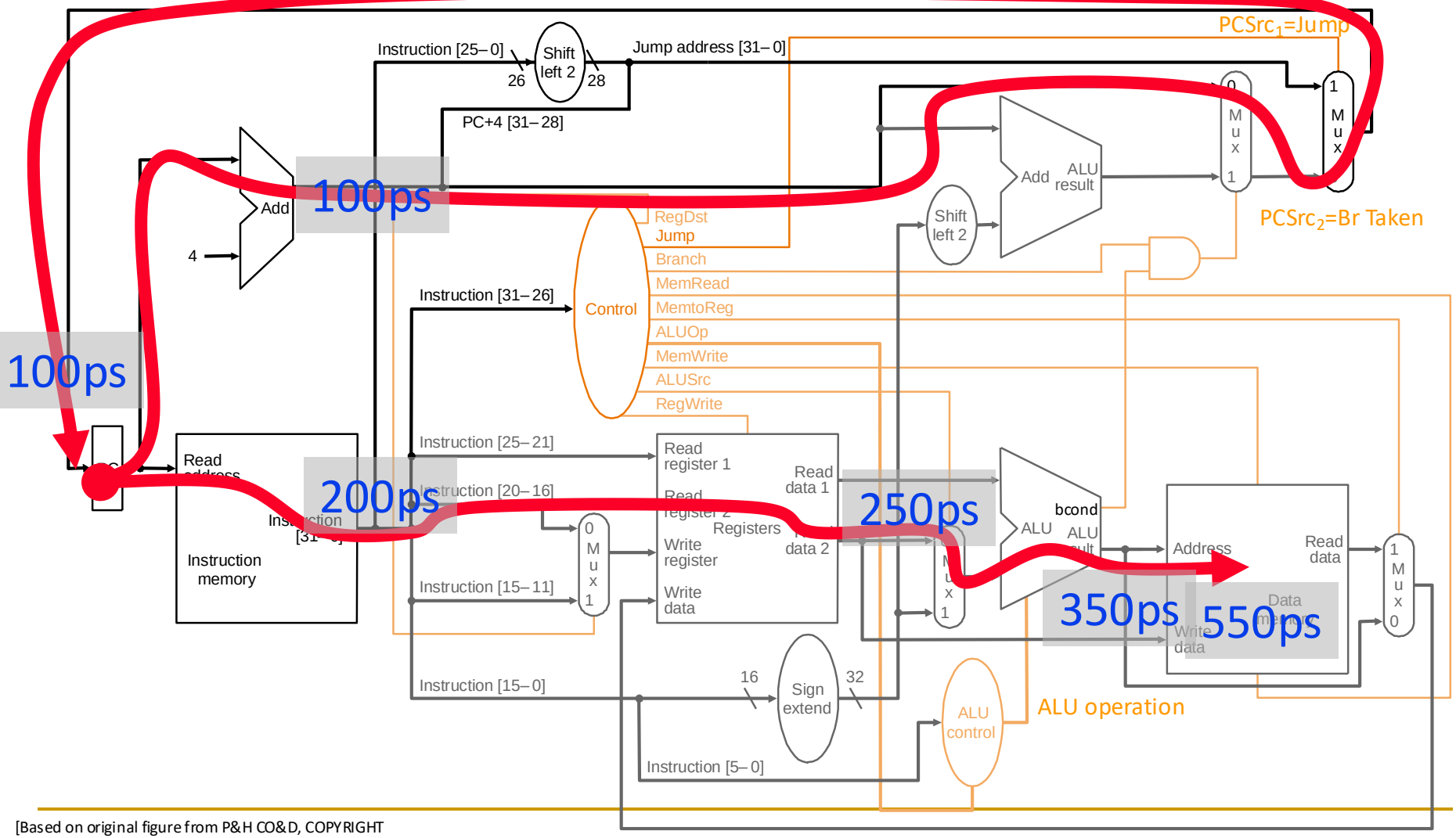
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LW



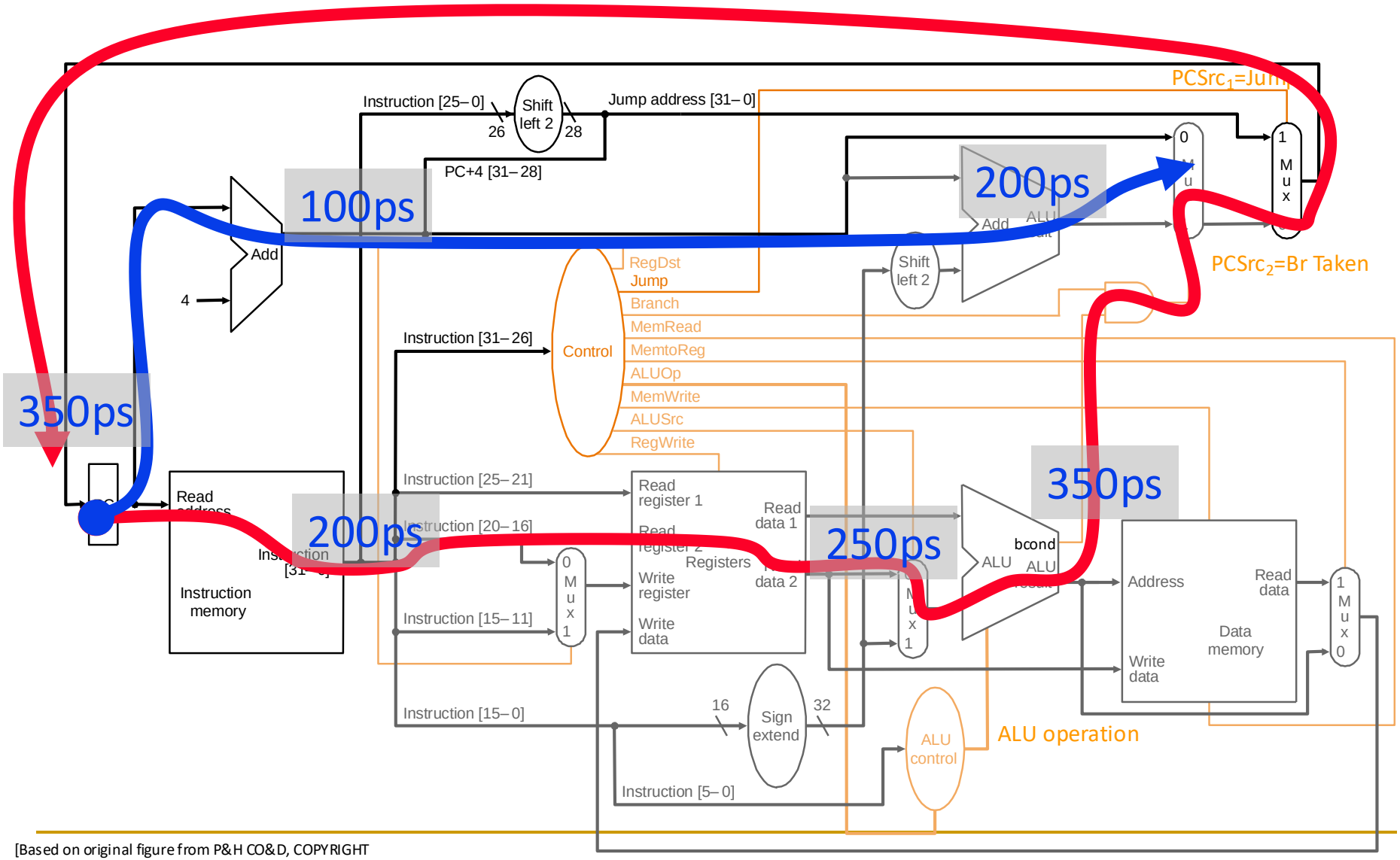
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SW



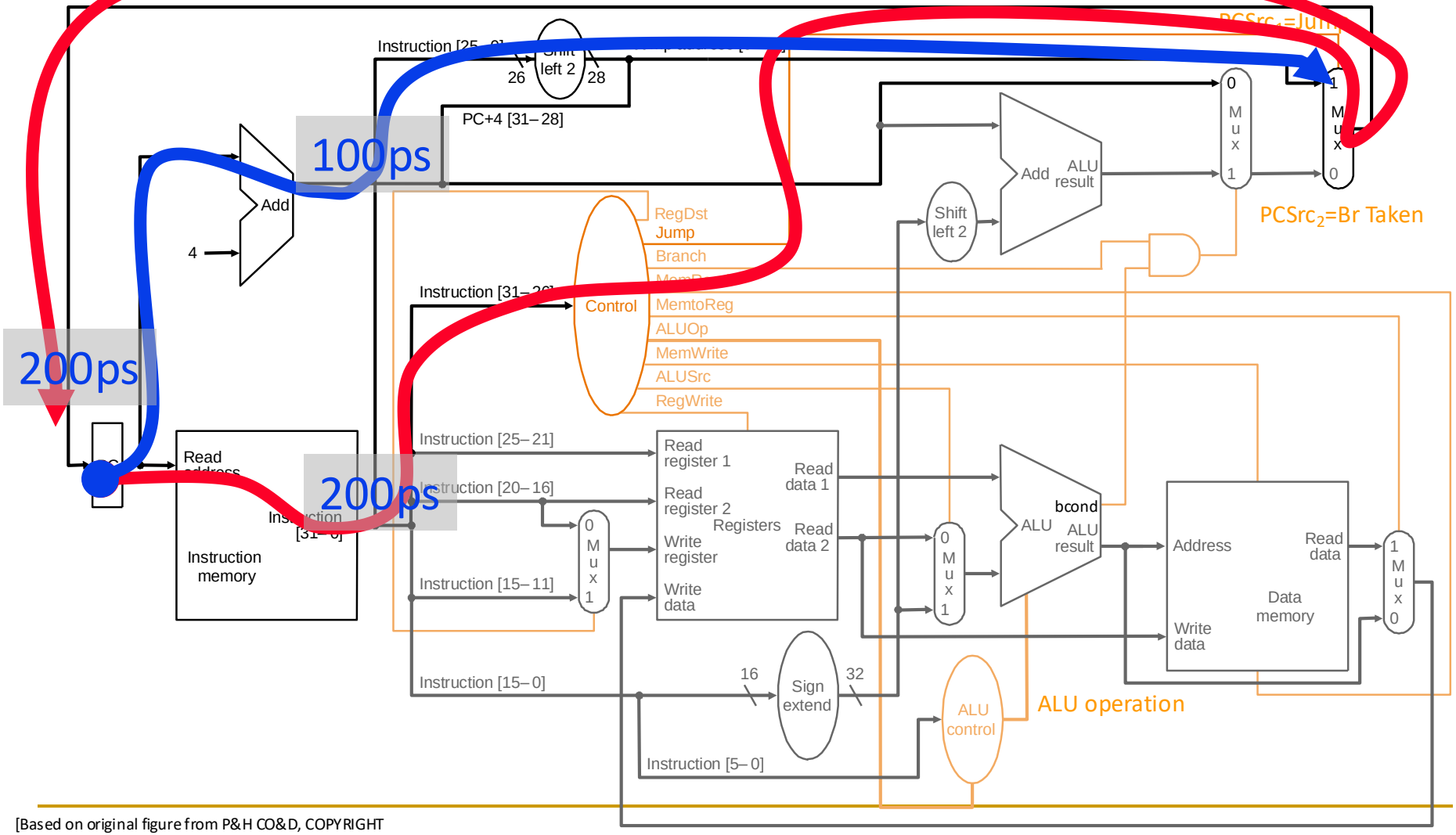
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Branch Taken



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Jump



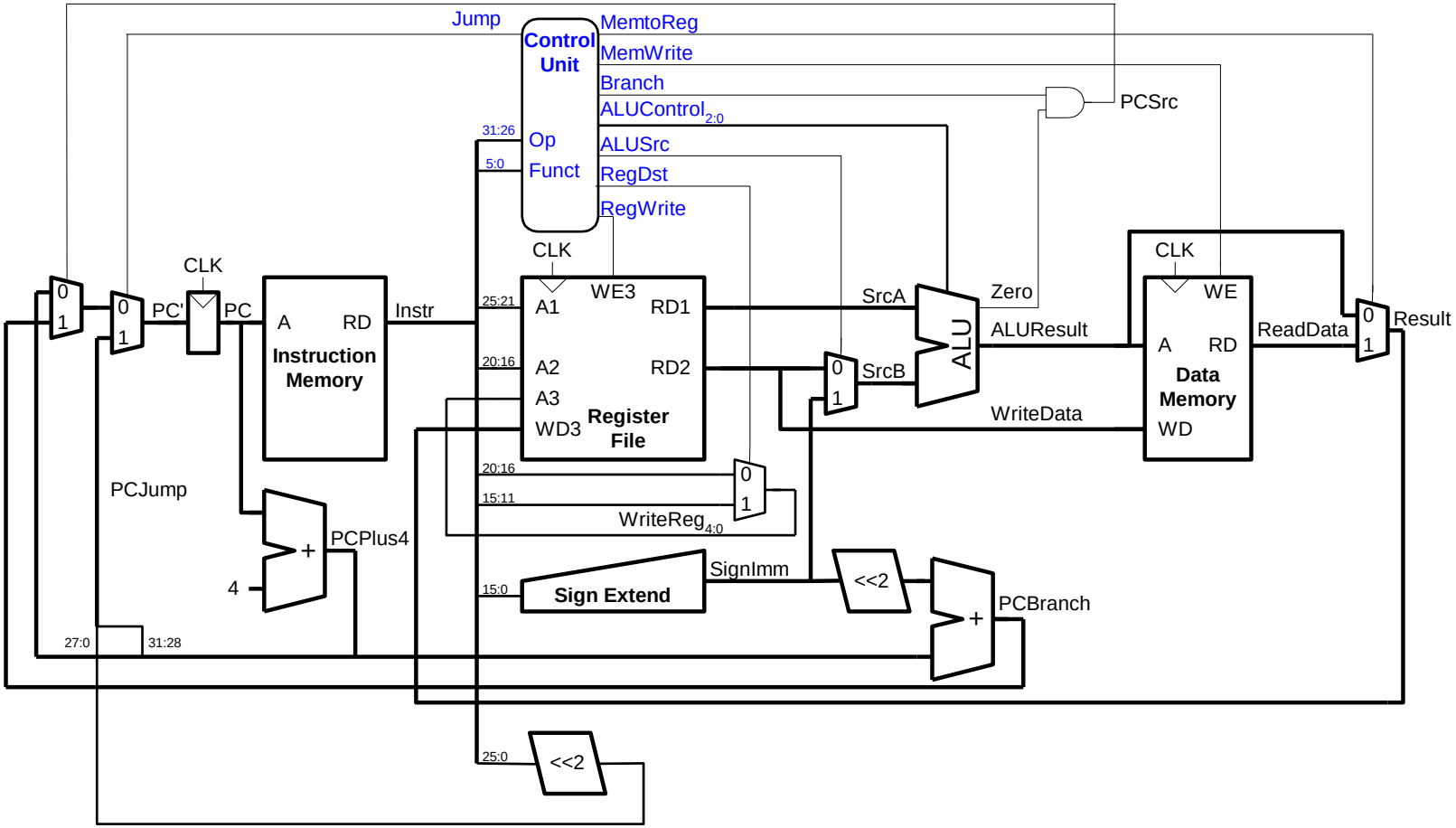
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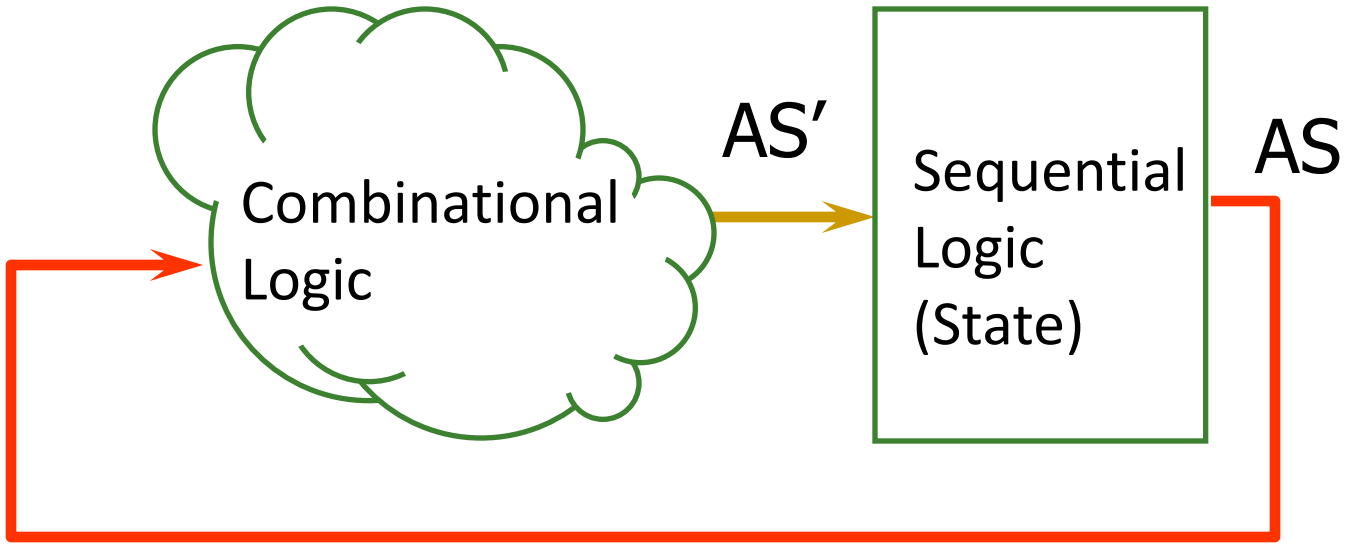
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Branch	200	50	100			350
Jump	200					200

Review: Single-Cycle MIPS Microarchitecture



Review: Single-Cycle MIPS FSM

- Single-cycle machine



AS: Architectural State